

The Price of Risk: Statistical Learning Approaches to Loan Interest Rate Prediction

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1 Introduction

When a person applies for a personal loan on LendingClub, they receive an interest rate that reflects the LendingClub platform’s assessment of their credit risk. This rate is not arbitrary; it encodes a judgment about the probability of default, the loan term, and the borrower’s financial behavior. Understanding what drives that rate and how well a statistical model can reproduce it from public information is both practically important and statistically interesting.

Research Questions: This project investigates three questions.

1. How well do publicly observable borrower financials explain LendingClub’s assigned interest rates, and which financial signals carry the most pricing power?
2. To what extent do unsupervised borrower clusters correspond to LendingClub’s risk tiers, and what financial profiles define each segment?
3. At what point does model complexity stop yielding meaningful accuracy gains, and what does this imply for model selection in a regulated lending context?

Algorithmic lending decisions affect millions of borrowers, and regulators are increasingly demanding explainability. A model that achieves 98% of a black box ensemble’s accuracy while remaining fully auditable may become the model that can legally be deployed.

2 Data Description

The dataset is the **LendingClub Loan Data 2007–2020**, taken from Kaggle.¹ The full set contains about 2.9 million loans. We took a stratified sample of 50,000 loans, using deciles of the target variable (`int_rate`, the annual interest rate in percent) to maintain the distribution. The response is continuous and ranges from about 5% to 31%.

Preprocessing: The variable `grade` was removed because LendingClub derives it directly from `int_rate`. Keeping it would be a form of target leakage. Administrative identifiers and columns with more than 30% missing values were also dropped. Numeric variables were winsorised at the 1st and 99th percentiles to limit the influence of extreme outliers. We engineered a new feature, `income_to_loan` (annual income divided by loan amount), to capture affordability. After one hot encoding the categorical variables (`home_ownership`, `purpose`, `term`), we applied a VIF (variance inflation factor) filter on the numeric predictors. This kept 13 base features, which expanded to 19 encoded features for modelling. Key predictors include: `loan_amnt`, `annual_inc`, `dti`, `fico_range_low`, `revol_util`, `emp_length`, `home_ownership`, `purpose`, `open_acc`, `delinq_2yrs`, `pub_rec`, `term`, `mort_acc`, `installment`, and others.

¹<https://www.kaggle.com/datasets/ethon0426/lendingclub-20072020q1>

3 Preliminary Studies: EDA and Dimensionality Reduction

The response variable has a mean of 13.0% and a right skew of 0.81. This means most borrowers pay between 10% and 15%, but a long tail goes up to more than 25%. Figure 1 put together four main findings from the exploratory analysis.

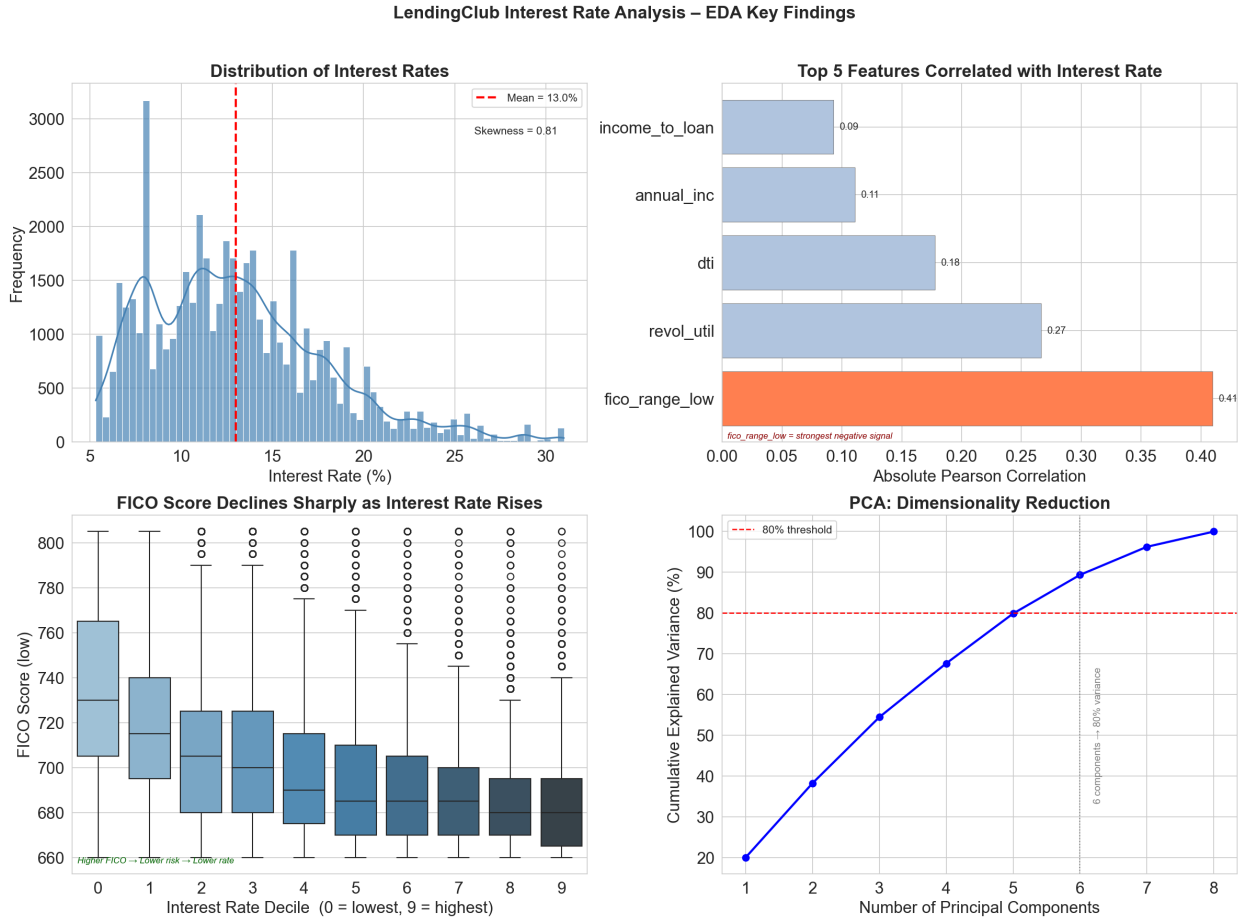


Figure 1: Summary of exploratory data analysis.

The top right panel of Figure 1 shows that `fico_range_low` has the strongest linear correlation with `int_rate` ($r = -0.41$). That is a negative relationship: higher FICO scores are linked to lower interest rates. The next strongest correlation is `revol_util` ($r = +0.27$), followed by `dti` ($r = +0.18$) and `annual_inc` ($r = -0.11$). Income matters, but its effect is much smaller than credit behaviour. The bottom left panel plots median FICO scores against interest rate deciles. As rates go up, FICO scores drop in a clear, steady way. This visual confirms the correlation finding. The bottom right panel shows a PCA scree plot. Six principal components are needed to explain 80% of the variance. That tells us that credit risk is not a single dimension; it is a mix of several underlying factors. This is why non linear models may be useful.

Figure 2 presents the distributions of key features and their correlations. `annual_inc` is heavily right skewed (skew = 1.83). Many borrowers have moderate incomes, but a few earn much more. `revol_util` is nearly symmetric (skew = 0.04), which is convenient for regression. The correlation matrix in the lower triangle reveals a moderately strong negative relationship between `fico_range_low` and `revol_util` ($r = -0.50$). Borrowers who use a large fraction of their revolving credit tend to have lower FICO scores. Even with this overlap, both features survive the VIF(Variance Inflation Factor) filter because they provide complementary risk information.

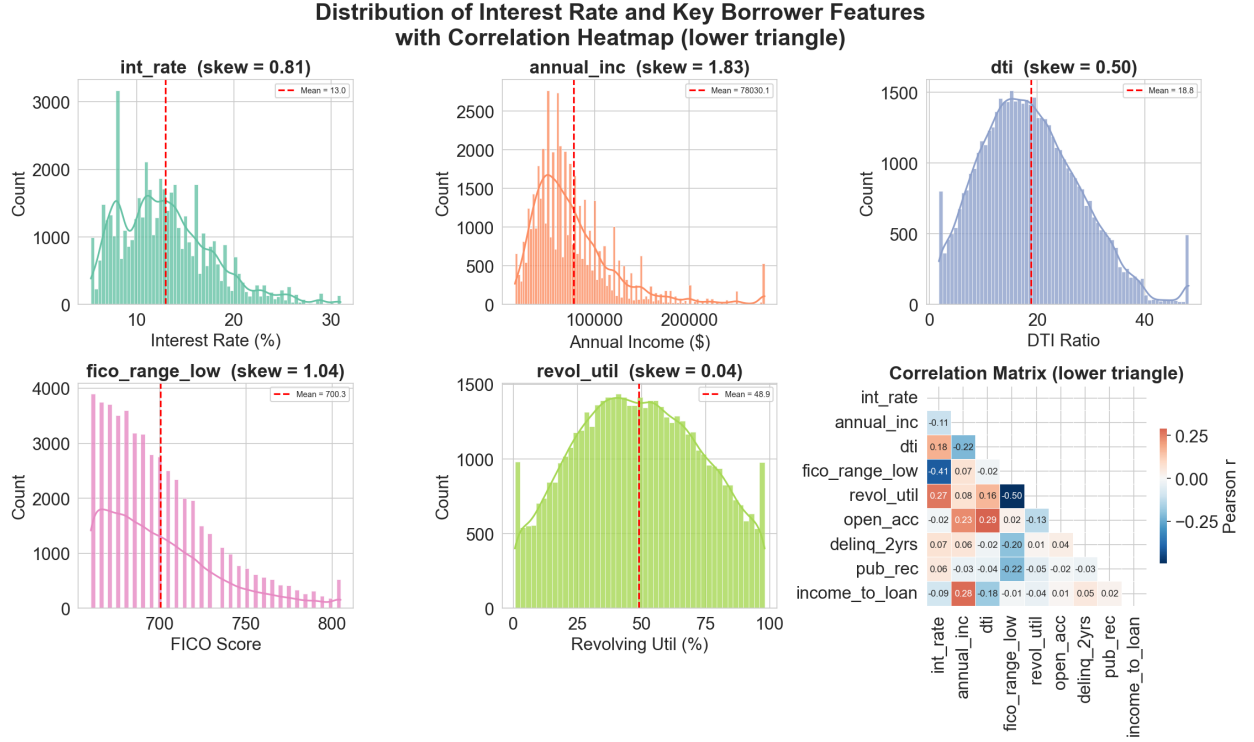


Figure 2: Distributions of five main features and their pairwise correlations.

4 Statistical Analysis

4.1 Unsupervised Learning: Borrower Segmentation (RQ2)

Before building any predictive model, we asked whether the data naturally splits into borrower risk segments. Importantly, clustering was done without using `int_rate`. We only looked at the rates afterward to interpret the segments.

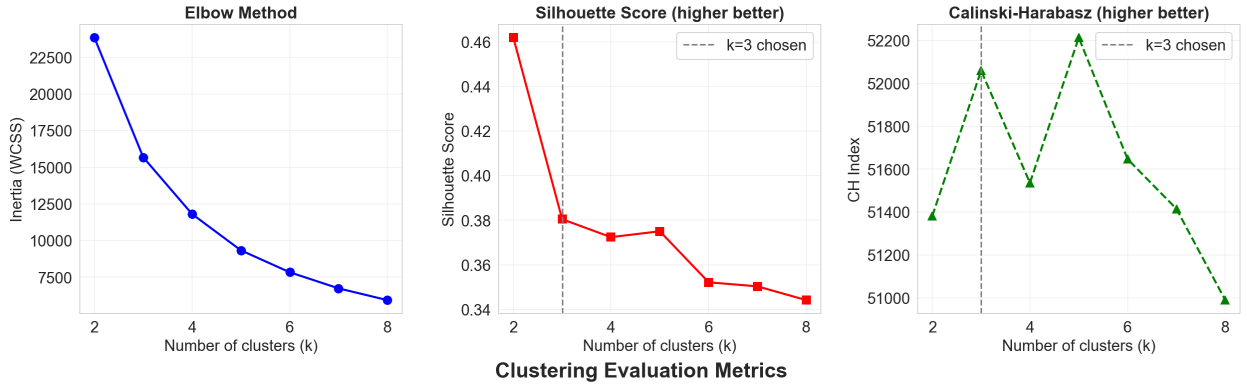


Figure 3: Three metrics for choosing the number of clusters.

Choosing the number of clusters: Figure 3 shows three evaluation metrics for $k = 2$ through 8. The silhouette score peaks at $k = 2$ (score = 0.46), meaning the strongest split in the data is binary, but the values are trying to elbow out beyond $k = 3$. However, the Calinski Harabasz index supports $k = 3$. Because $k = 3$ maps directly to the industry standard categories of *Prime*, *Near Prime*, and *Subprime*, we chose three clusters. It balances statistical evidence with real world interpretability.

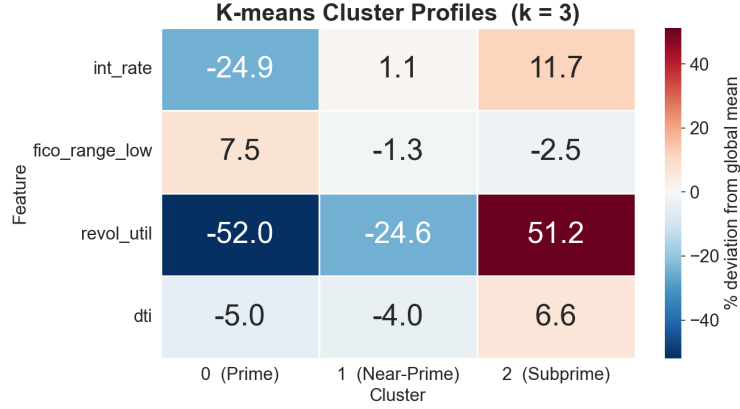


Figure 4: Cluster profiles as percent deviation from global means.

Cluster profiles: The heatmap in Figure 4 shows how each cluster differs from the global average. The Prime cluster has revolving utilisation 52% below the average and FICO score 7.5% above the average. The observed average rate for this group is 9.8%. The Near Prime cluster stays close to the global mean on all features except revolving utilisation, which is 24.6% below the average, with an average rate of 13.1%. The Subprime cluster is striking: its revolving utilisation is 51.2% above the global mean, but its FICO score is only 2.5% below the average. Its average rate is 14.5%. This shows that the subprime premium is driven much more by credit card usage behaviour than by a low FICO score.

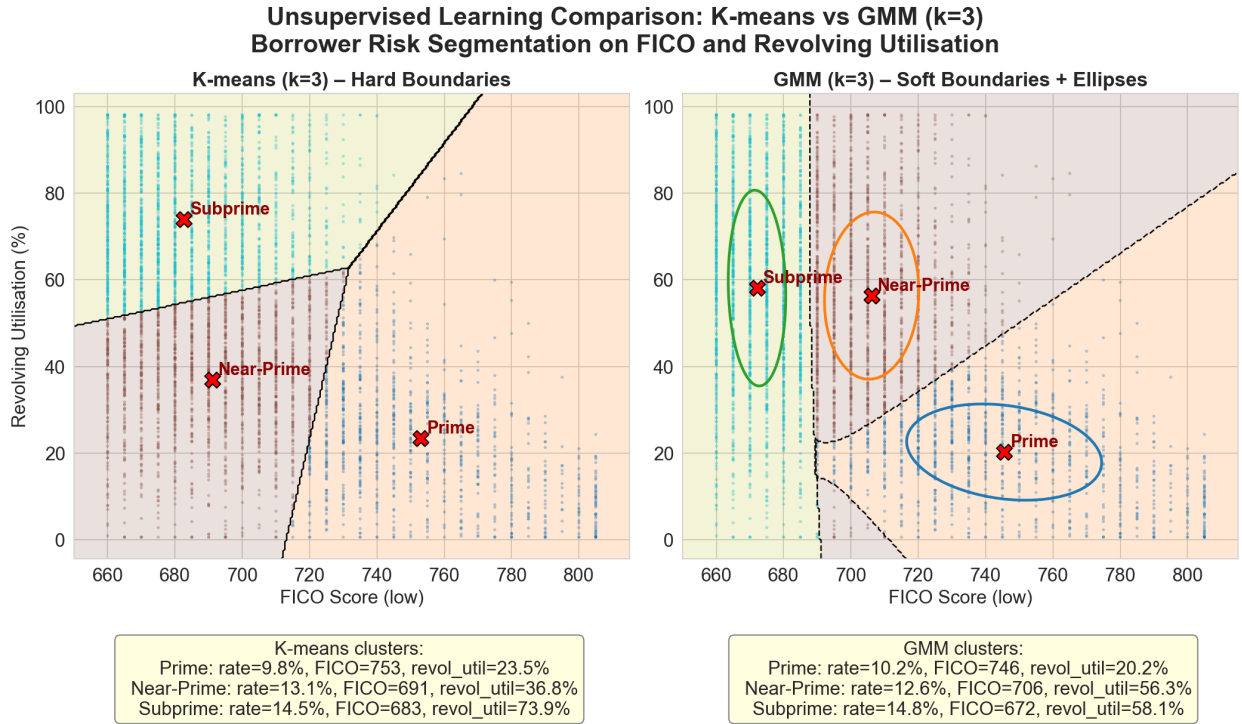


Figure 5: K-means (hard boundaries) versus GMM (soft ellipses) on FICO and revolving utilisation.

Robustness check with GMM: We also fitted a Gaussian Mixture Model with $k = 3$. GMM gives soft cluster boundaries instead of the hard boundaries of K-means. Figure 5 compares the two methods. Both recover the same three segments, and their average interest rates agree within 0.4 percentage points. The GMM ellipses show that the Prime cluster is stretched along the FICO axis, while the Subprime cluster stretches along the utilisation axis. Two very different algorithms reaching the same conclusion give us confidence that the three segments are truly present in the data.

4.2 Regression (RQ1 and RQ3 Baseline)

We fitted four linear models on an 80/20 train test split: ordinary least squares (OLS), Ridge (L2), Lasso (L1), and ElasticNet. Hyperparameters were chosen by five fold cross validation. Ridge selected $\alpha \approx 126.4$, which shrinks all coefficients but keeps every feature. Lasso selected $\alpha = 0.0079$ and set six coefficients to exactly zero. Those zeroed features were all purpose and home ownership dummy variables. This suggests that once you include numeric risk signals like FICO and utilisation, those categorical variables add almost no extra information. ElasticNet's optimal ℓ_1 ratio went to 1.0, so it collapsed to Lasso; the L2 penalty did not help.

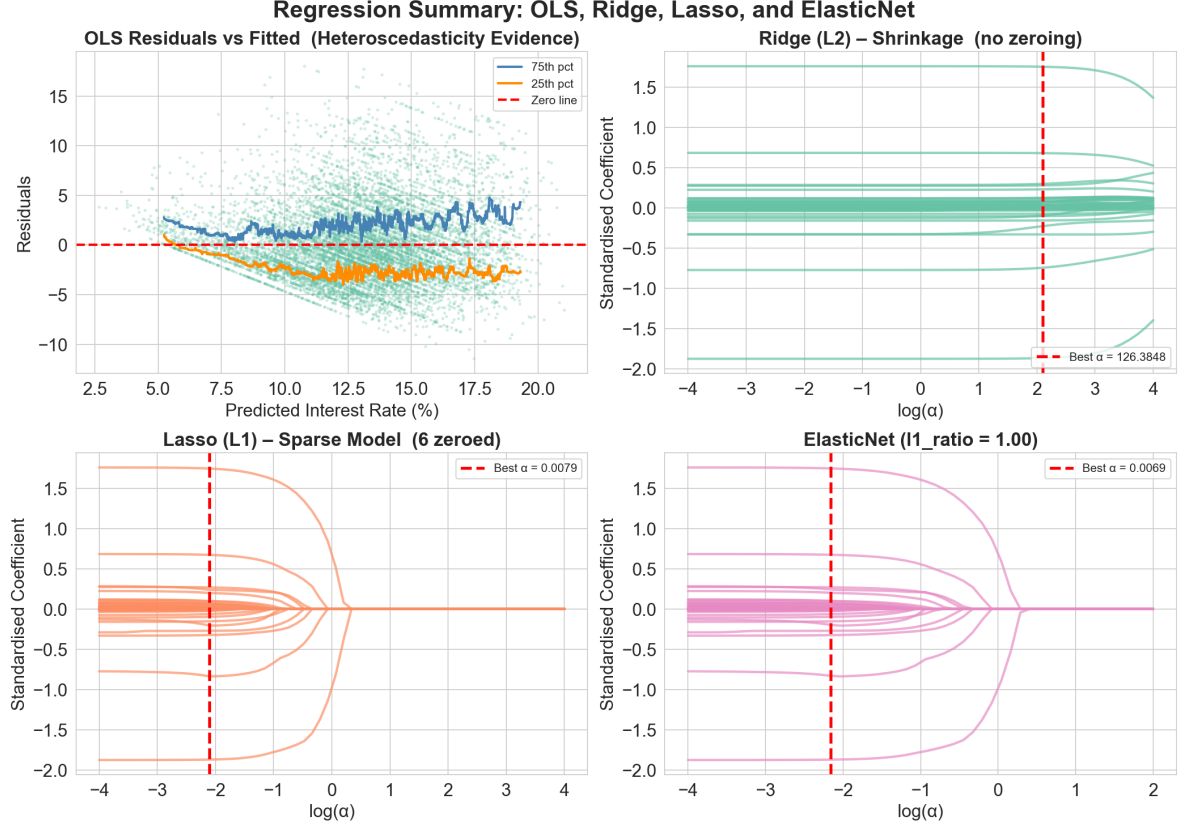


Figure 6: OLS residual plot and regularisation paths.

Table 1 gives test set performance. All four linear models have nearly identical RMSE (about 3.86) and R^2 (about 0.367). This is an important clue. Regularisation changes the coefficient values but cannot recover information that OLS missed. The bottleneck is not overfitting or collinearity; it is the absence of non-linear structure in the features. The R^2 ceiling of 0.37 motivates us to try non-linear models.

Table 1: Test set performance of linear models.

Model	RMSE	MAE	R^2
OLS	3.8584	2.9648	0.3669
Ridge	3.8583	2.9646	0.3669
Lasso	3.8588	2.9648	0.3667
ElasticNet	3.8587	2.9648	0.3668

Figure 7 shows the absolute coefficients from ElasticNet. `fico_range_low` is clearly the most important predictor. The second most important is `term_60months`. A 60 month loan term is a product feature, not a borrower risk characteristic; lenders charge a premium for longer terms regardless of borrower quality. The presence of `dti`, `revol_util`, and `annual_inc` in the top ten confirms the EDA finding: credit behaviour matters more than raw income.

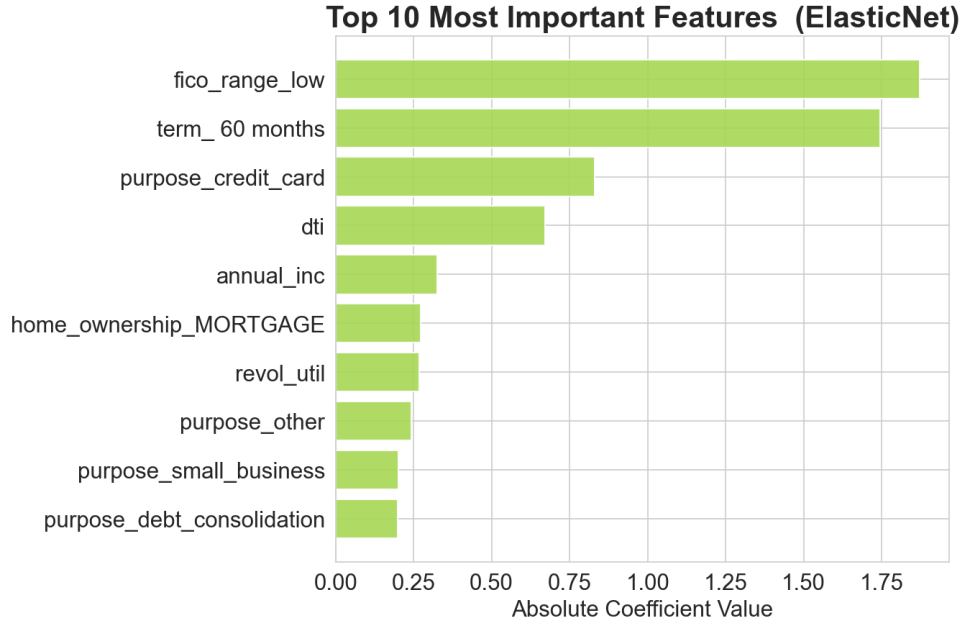


Figure 7: Top ten features by absolute coefficient in ElasticNet.

The OLS residual plot (top left of Figure 6) shows a fan shape. The spread of residuals grows as the predicted interest rate rises. This is called heteroscedasticity, meaning the error variance is not constant. The model is more uncertain about high rate loans.

4.3 Random Forest: Non-linear Interactions (RQ1 and RQ3)

We tuned a Random Forest over a grid of tree counts and maximum depths using three-fold cross-validation. The best model used 200 trees and maximum depth 10. It achieved a test RMSE of **3.8109** and $R^2 = 0.382$, improving on Ridge by 0.047 percentage points. The R^2 gain of 1.5 percentage points over the linear models shows that the ensemble captures real non-linear structure, though the improvement is modest.

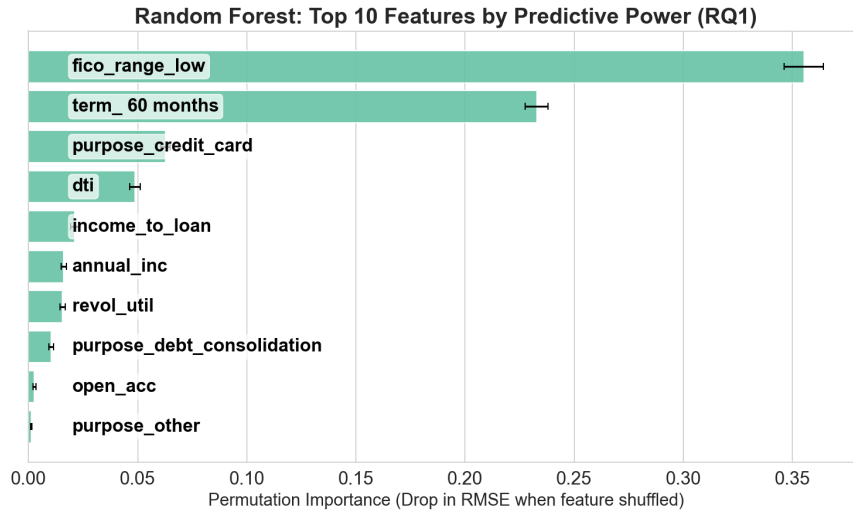
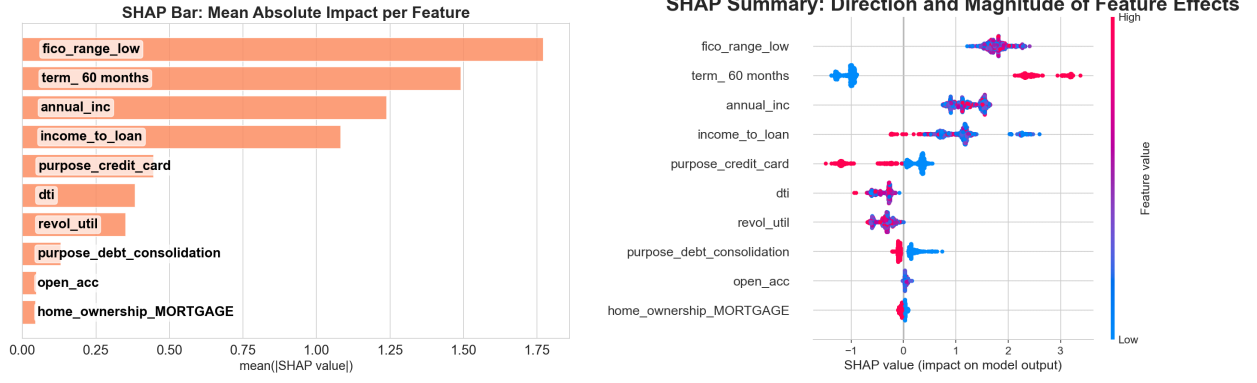


Figure 8: Permutation importance for the top 10 features (Random Forest).

Feature importance: We measured importance by two complementary methods. Permutation importance (Figure 8) shuffles each feature’s values and records the RMSE increase; a larger drop means greater importance. **fico_range_low** scores 0.35, roughly 50% larger than the second-ranked **term_60months** (0.23), with a sharp drop to 0.06 for all remaining features.

The SHAP analysis (Figure 9) independently confirms this ranking: both methods place `fico_range_low` first and `term_60months` second, making this a method-independent answer to RQ1. The beeswarm plot reveals the direction of each effect. Sixty-month loans receive a positive SHAP push of up to +3.5 percentage points — a structural term premium unrelated to borrower risk. Credit-card purpose loans receive a negative push of around -1 pp, reflecting a product-level discount. High revolving utilisation and high DTI push predictions upward, consistent with the EDA correlations but now quantified at the individual loan level.



(a) Mean $|\text{SHAP}|$ per feature.

(b) Direction and magnitude of SHAP effects.

Figure 9: SHAP analysis. Red dots indicate high feature values; blue indicates low.

Partial dependence: The one-dimensional PDPs (Partial Dependence Plots) for `fico_range_low`, `revol_util`, and `dti` (Figure 11) are all nearly flat across their respective standardised ranges. `fico_range_low` sits at a constant partial dependence of approximately 15.2%; `revol_util` at 13.1% with a marginal decrease of 0.1 percentage points at high utilisation; and `dti` at 12.9% with no big variation. This flatness does not contradict the high permutation and SHAP importance of these features. Rather, it indicates that the Random Forest captures their effect primarily through *interactions* with other features, particularly `term_60months` and loan purpose, rather than through independent additive main effects. Marginalising over the full joint distribution of all other features cancels out these interaction contributions, producing flat marginal curves even for the model’s most important predictors.

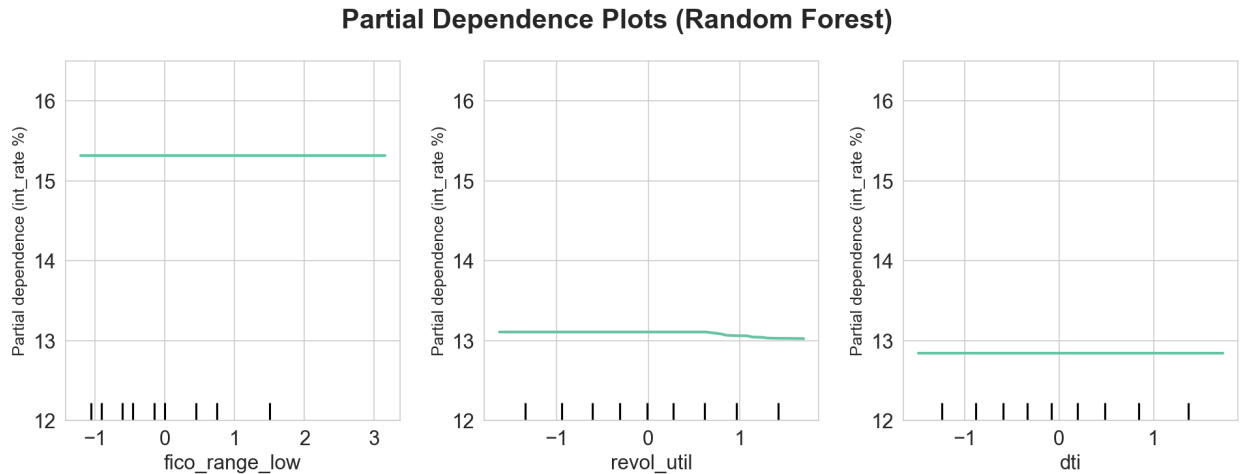


Figure 10: 1-D PDPs for the top three numeric features.

Learning curve and the feature ceiling (RQ3): Figure 11 shows the cross-validation RMSE falling from 3.95 at 3,000 training examples to approximately 3.84 at 15,000 examples, after which it flattens. Adding the remaining 17,000 examples reduces RMSE by less than 0.01 percentage points.

This plateau directly answers RQ3: the accuracy ceiling comes from the features, not from insufficient data or model capacity. The private bureau data that LendingClub uses in production includes full tradeline histories, payment records, and enquiry counts. These variables are simply not present in this public dataset.

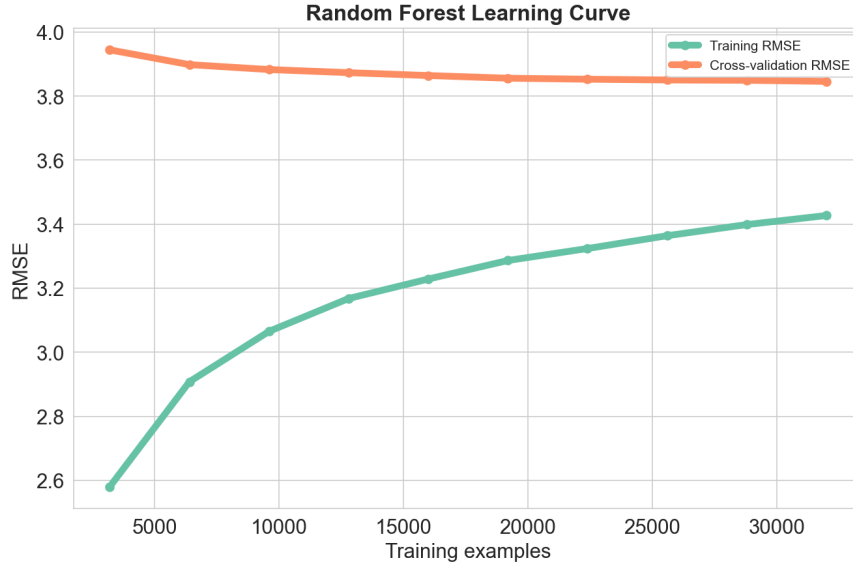


Figure 11: Random Forest learning curve showing plateau at $\sim 15,000$ examples.

4.4 Neural Network: Multi-Layer Perceptron MLP (RQ3)

We built a multilayer perceptron (MLP) in PyTorch.

The architecture was: Input (38 features) $\rightarrow$ Linear(128) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(64) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(1). We used the Adam optimiser with a learning rate of 0.001 and weight decay of 10^{-4} . Training stopped early after 22 epochs because the validation loss did not improve for five epochs.

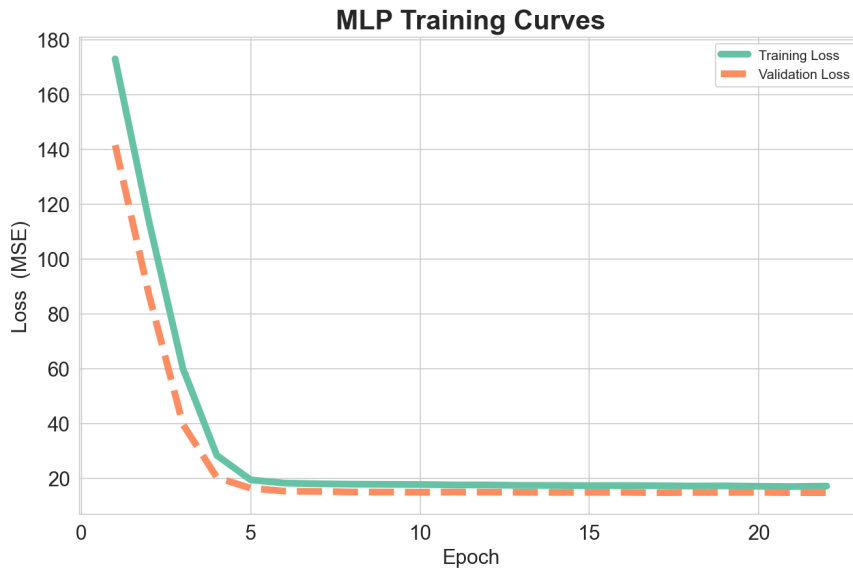


Figure 12: Training and validation loss curves for the MLP.

The loss curves in Figure 12 show fast convergence and no signs of overfitting. Validation loss stays below training loss throughout because Dropout is active only during training. The final test RMSE was 3.8337, placing the MLP between Random Forest and the linear models. Hence, tree based ensembles usually outperform neural networks when the feature set is moderate.

4.5 Final Model Comparison (RQ3)

Figure 13 and Table 2 compare all six models on the test set. The total range of RMSE values is only 0.048 percentage points, from the best (Random Forest, 3.8109) to the worst (Lasso, 3.8588). That is an extremely narrow spread, especially given the variety of models.

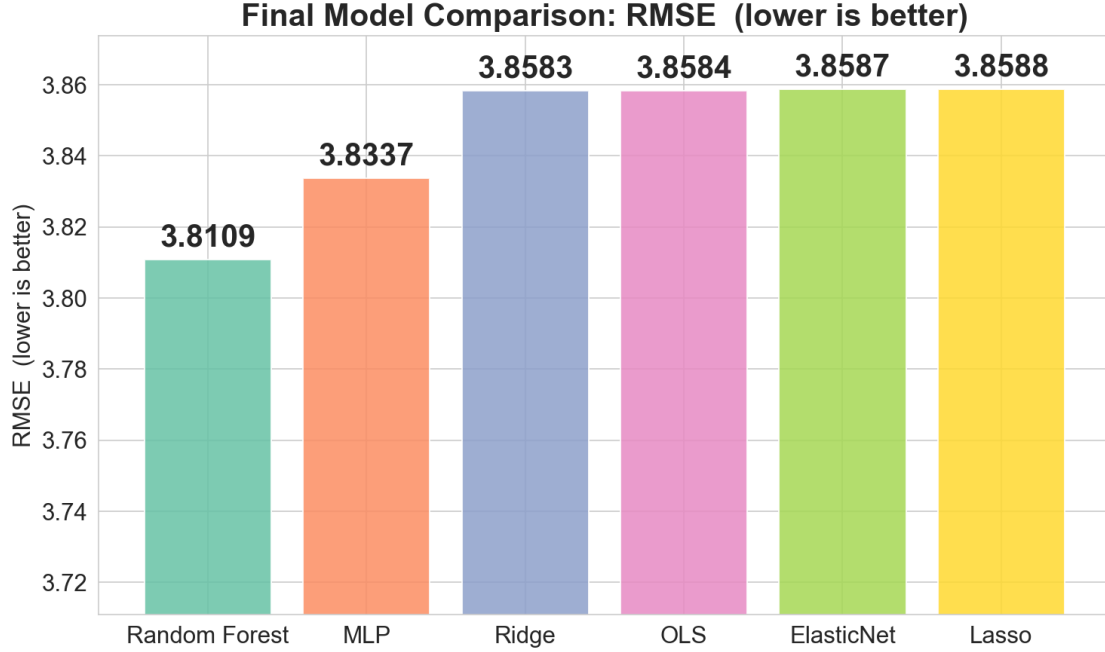


Figure 13: RMSE comparison across all six models.

Table 2: Final test set performance of all models.

Model	RMSE	MAE	R^2	Rank
Random Forest	3.8109	2.9126	0.3824	1st
MLP	3.8337	2.8937	0.3749	2nd
Ridge	3.8583	2.9646	0.3669	3rd
OLS	3.8584	2.9648	0.3669	4th
ElasticNet	3.8587	2.9648	0.3668	5th
Lasso	3.8588	2.9648	0.3667	6th

The narrow range answers RQ3 directly. Adding model complexity yields only a tiny improvement in accuracy. The Random Forest reduces RMSE by 0.05 relative to Ridge, and the MLP sits in between. For a regulated lending context, this has a clear practical meaning. Ridge regression gives 98.8% of the Random Forest’s accuracy but offers full interpretability. Each coefficient can be examined, explained, and audited. Under fair lending laws and Basel III model risk rules, a black box model is very difficult to justify when a nearly as accurate linear model exists.

5 Conclusions

We applied a range of statistical learning tools, including *exploratory data analysis*, *K-means*, *Gaussian mixture clustering*, *regularized regression* (*Ridge*, *Lasso*, *ElasticNet*), *random forests* with permutation importance and *SHAP values*, and a *multilayer perceptron neural network* to predict LendingClub interest rates.

The three research questions are answered as follows.

RQ1 (pricing power): FICO score (`fico_range_low`) is the single most important predictor. This result was consistent across four different methods: Pearson correlation, ElasticNet coefficients, Random Forest permutation importance, and SHAP values. Credit behaviour, especially revolving utilisation, matters more than raw income. The 13 public features together explain only about 37% of the variance in interest rates. LendingClub almost certainly uses proprietary bureau data (full tradeline histories, alternative credit signals) that is not available in this public dataset.

RQ2 (natural risk segments): K means with $k = 3$, independently validated by GMM, recovers three borrower tiers that match the Prime, Near Prime and Subprime classification. The average rates for these clusters are 10%, 12.9%, and 14.7%. The Subprime segment is distinguished primarily by very high revolving utilisation (+51% above the mean), not by a very low FICO score. Behavioural signals matter more than the summary credit score for identifying the riskiest borrowers.

RQ3 (complexity versus accuracy): All six models have a test RMSE range of only 0.048 (absolute), from 3.8588 down to 3.8109. The Random Forest learning curve plateaus at 15,000 training examples, showing that the limit is the feature information, not the amount of data or the model’s capacity.

For a regulated lender, Ridge regression is the recommended model. It achieves 98.8% of the best model’s accuracy with less computational cost and is fully interpretable.

When a bank or lending company has to follow strict rules, they need to be able to explain exactly why a borrower got a certain interest rate. A simple model like Ridge regression is almost as accurate as a complex model, and it is much easier to explain. The very small gain in accuracy from using a Random Forest or a neural network is not worth the trouble of not being able to fully explain their decisions.

Limitations: The OLS residuals show clear heteroscedasticity, which is a common pattern in interest rate data. This finding suggests that weighted least squares or quantile regression could be applied in future work to better model the changing variance across the rate spectrum. Breaking the $R^2 \approx 0.37$ ceiling would require richer data, such as cash flow history, rent payments, or full bureau tradelines.

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